

SET THREE

SECTION A [40 MARKS]

Attempt **ALL** the Questions in this Section.

Question 1

- a) Define the term principles of programming languages. **(2 marks)**
- b) State any **two** objectives of principles of programming languages. **(2 marks)**
- c) With examples of languages in each, state any **three** programming paradigms. **(6 marks)**
- d) Explain why there is need of so many programming paradigms. **(4 marks)**
- e) Explain any three skills required to become a computer programmer. **(6 marks)**
- f) State the three types of computer programming languages. **(3 marks)**
- g) State any three advantages of learning computer programming. **(3 marks)**
- h). State the four of computer programming languages. **(4 marks)**
- i) Explain the following as characteristics of a successful computer programming language: **(2 marks @)**
 - i) Support
 - ii) Simplicity and Readability
 - iii) Reliability
- j) Programming languages support different data types. These usually include STRING and REAL. Complete the table by giving four other data types and an example data value for each. **(4 marks)**

Data type	Example data value

SECTION B [60 MARKS]

Attempt **ONLY THREE** Questions in this Section.

Question 2

- a) Define an algorithm. **(2 marks)**
- b) Algorithms for solving computational tasks is divided into two sections, explain them. **(4 marks)**
- c) Explain any **three** pseudo-code key words. **(6 marks)**
- d) State and draw the corresponding flow-chart symbols of the key-word in (c) above. **(3 marks)**
- e) Draw a flow chart solution to the problem below: **(5 marks)**

You are required to capture two integers from the key board and the computer should display the bigger number of the two.

Question 3

- a) Define the following terms: **(2 marks @)**
 - i) Data
 - ii) Data item
 - iii) Record
 - iv) File
 - v) Data structure
- b) State any **three** rules of naming data items. **(3 marks)**
- c) List two examples of wrong data item names **(2 marks)**
- d) State any five operations on data structures. **(5 marks)**

Question 4

- a) Define the term code translation. **(2 marks)**
- b) Explain why translation is important. **(2 marks)**
- c) State any two differences between compilation and interpretation. **(4 marks)**

d) Explain the six stages of compiling a program below: **(2 marks @)**

- i) Lexical analysis.
- ii) Symbol table construction.
- iii) Syntax analysis.
- iv) Semantic analysis.
- v) Code generation.
- vi) Optimization.

Question 5

a) Define the following terms: **(2 marks @)**

- i) Statement
- ii) Block
- iii) Expression
- iv) Operator
- v) Control flow

b) With examples, explain the following operators: **(2 marks @)**

- i) Comparison
- ii) Mathematical
- iii) Assignment

c) Explain the flow control statements below: **(2 marks @)**

- i) Selection
- ii) Iteration

Question 6

a) Define the following Object Oriented terms: **(2 marks @)**

- i) Class

ii) Object

iii) Encapsulation

iv) Decomposition

v) Aggregation

c) With examples, explain the term modularity. **(6 marks)**

D) State three differences between object-oriented paradigm and imperative paradigm **(4 marks)**

END